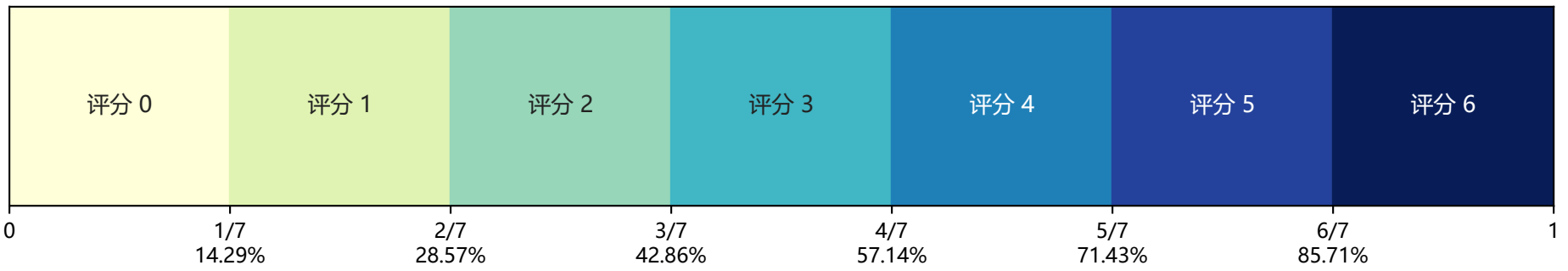
分级阈值：第 1 次重复、第 1 个外层训练折（n=200）
只用训练标签计算，边界相等归低档；每一折的全部阈值另存 CSV

0–6 七档对应的分位概率区间（并列值保留，实际人数不必完全相等）



	Q1/7 (14.29%)	Q2/7 (28.57%)	Q3/7 (42.86%)	Q4/7 (57.14%)	Q5/7 (71.43%)	Q6/7 (85.71%)
A_malt_all	-0.643	-0.426	-0.207	0.069	0.431	0.763
A_malt_grain	-0.400	-0.232	-0.114	0.003	0.180	0.503
A_malt_bread	-0.389	-0.255	-0.139	0.012	0.198	0.517
A_malt_cara	-0.521	-0.447	-0.255	-0.070	0.175	0.716
A_malt_burn	-0.372	-0.343	-0.271	-0.143	0.002	0.321
A_hops_all	-0.697	-0.458	-0.172	0.071	0.251	0.604
A_hops_citrus	-0.427	-0.356	-0.223	-0.087	0.124	0.399
A_hops_tropical	-0.393	-0.336	-0.240	-0.088	0.076	0.387
A_hops_noble	-0.567	-0.331	-0.186	-0.000	0.227	0.592
A_hops_woody	-0.324	-0.278	-0.174	-0.060	0.105	0.310
A_esters_all	-0.554	-0.298	-0.098	0.103	0.318	0.621
A_esters_ethac	-0.526	-0.346	-0.162	0.006	0.196	0.518
A_esters_isoaa	-0.485	-0.295	-0.142	0.048	0.309	0.600
A_esters_flower	-0.414	-0.348	-0.213	-0.056	0.114	0.448
A_ester_fruity	-0.425	-0.277	-0.131	-0.025	0.128	0.511
F_malt_all	-0.686	-0.456	-0.205	0.077	0.353	0.743
F_malt_grain	-0.441	-0.244	-0.110	-0.004	0.189	0.435
F_malt_bread	-0.458	-0.308	-0.189	0.032	0.263	0.511
F_malt_cara	-0.501	-0.382	-0.236	-0.048	0.226	0.669
F_malt_burn	-0.446	-0.375	-0.265	-0.170	0.036	0.505
F_hops_all	-0.742	-0.444	-0.166	0.069	0.419	0.745
F_hops_citrus	-0.411	-0.351	-0.202	-0.050	0.104	0.441
F_hops_tropical	-0.320	-0.303	-0.276	-0.124	0.109	0.409
F_hops_noble	-0.584	-0.409	-0.252	0.036	0.285	0.570
F_hops_woody	-0.356	-0.304	-0.193	-0.122	0.082	0.353
F_esters_all	-0.577	-0.294	-0.119	0.136	0.400	0.644
F_esters_ethac	-0.507	-0.317	-0.164	0.004	0.227	0.521
F_esters_isoaa	-0.461	-0.284	-0.130	0.013	0.332	0.606
F_esters_flower	-0.322	-0.286	-0.241	-0.064	0.089	0.335
F_ester_fruity	-0.388	-0.255	-0.159	-0.017	0.167	0.465
body	-0.582	-0.307	-0.130	0.094	0.363	0.609
bitternes	-0.811	-0.469	-0.185	0.068	0.326	0.636
co2	-0.602	-0.327	-0.145	0.101	0.321	0.681
aftertaste	-0.561	-0.289	-0.046	0.158	0.372	0.539
alcohol	-0.747	-0.385	-0.132	0.122	0.357	0.703
sweetness	-0.595	-0.357	-0.174	0.014	0.262	0.720
acidity	-0.492	-0.445	-0.365	-0.267	-0.061	0.562
dms	-0.501	-0.329	-0.201	-0.017	0.153	0.437
metallic	-0.528	-0.387	-0.225	-0.041	0.175	0.479
4vg	-0.477	-0.443	-0.318	-0.160	-0.004	0.508
t2n	-0.448	-0.316	-0.155	-0.033	0.110	0.324
diacetyl	-0.458	-0.352	-0.223	-0.009	0.177	0.502
stale_hops	-0.361	-0.345	-0.250	-0.159	0.032	0.307
orange	-0.258	-0.248	-0.233	-0.227	-0.085	0.194
coriander	-0.193	-0.180	-0.171	-0.166	-0.160	0.248
clove	-0.228	-0.220	-0.211	-0.203	-0.173	0.176
lactic	-0.325	-0.311	-0.306	-0.299	-0.273	0.766
acetic	-0.290	-0.280	-0.276	-0.267	-0.255	0.463
barnyard	-0.352	-0.340	-0.309	-0.170	0.034	0.336
overall	-0.577	-0.265	-0.095	0.106	0.293	0.507

